

Dynamical aspects of the effect of curiosity, reward incentives and their interaction on memory

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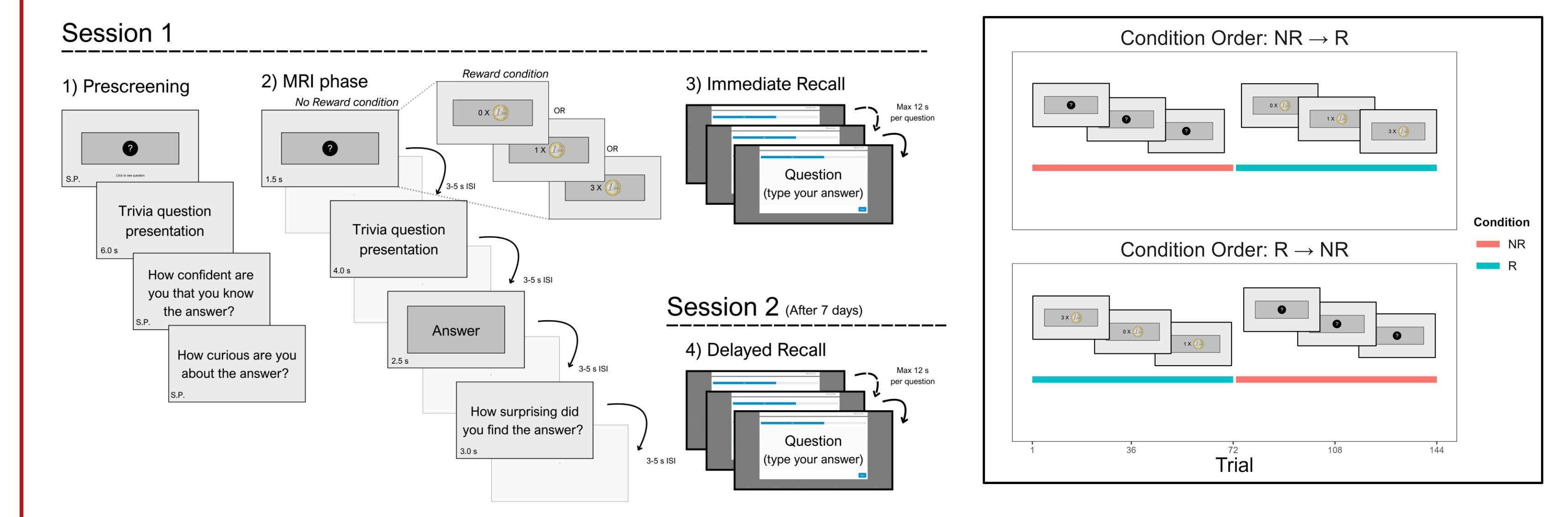
INTRODUCTION

- **Learning** happens because of **motivation**
- **Extrinsic** (e.g. reward incentives, grades, etc.) or **intrinsic** motivation (e.g. curiosity)
- In real world settings these **coexist**: but do they also **interact**?
- Duan et al. (2020) and Meliss et al. (2024): **additive** effect
- Murayama & Kuhbandner (2011) and Swirsky et al. (2021): **interaction** effect
- Resolve using an **adapted trivia paradigm** experiment

METHODS

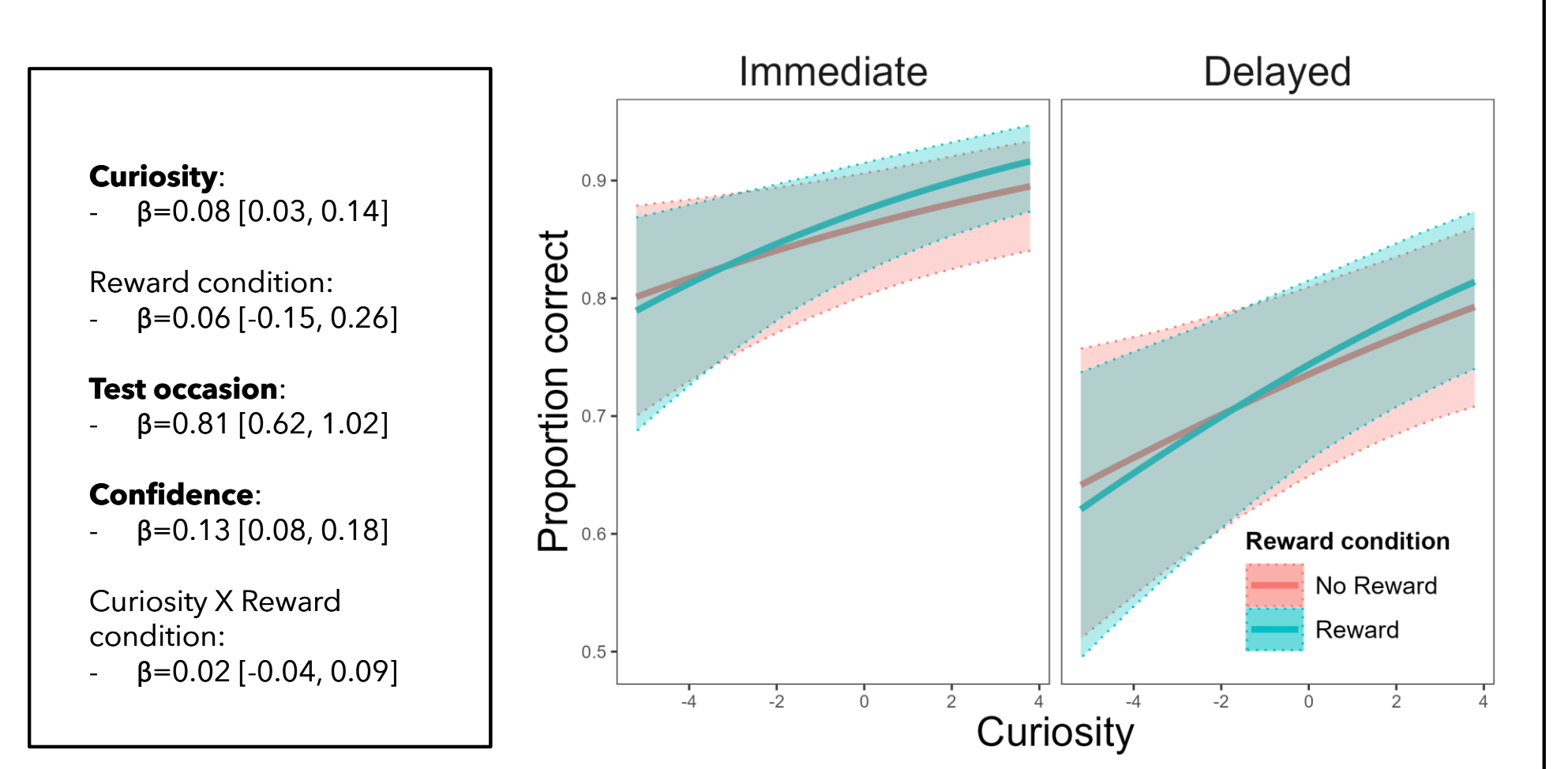
- **Preregistered** fMRI study
- Behavioural analysis using Bayesian Mixed Effects models (brms in R)
- Theoretically **maximal model** structure: **subject** and **item** intercepts and slopes for all within-subject effects

DESIGN

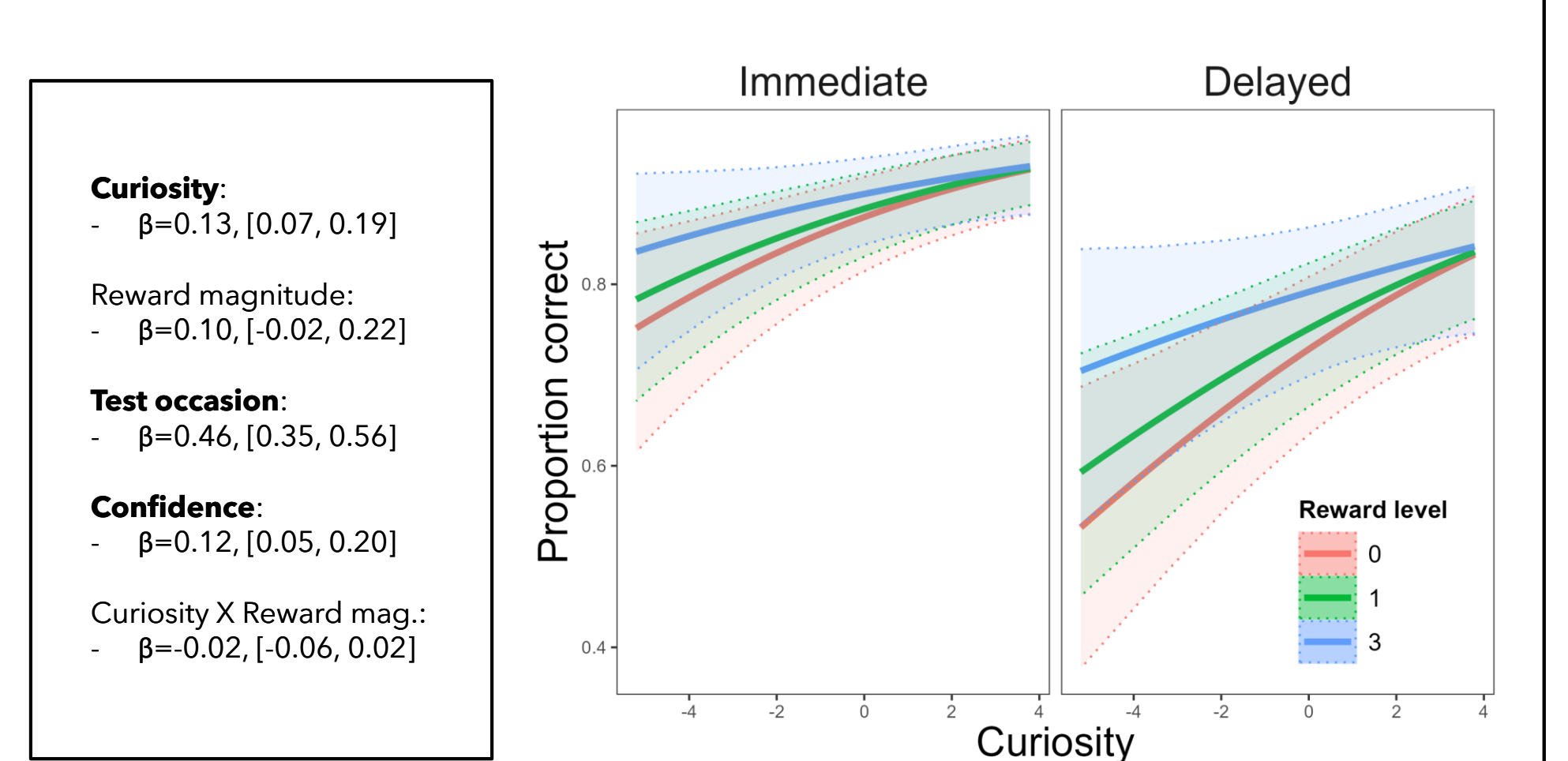


RESULTS

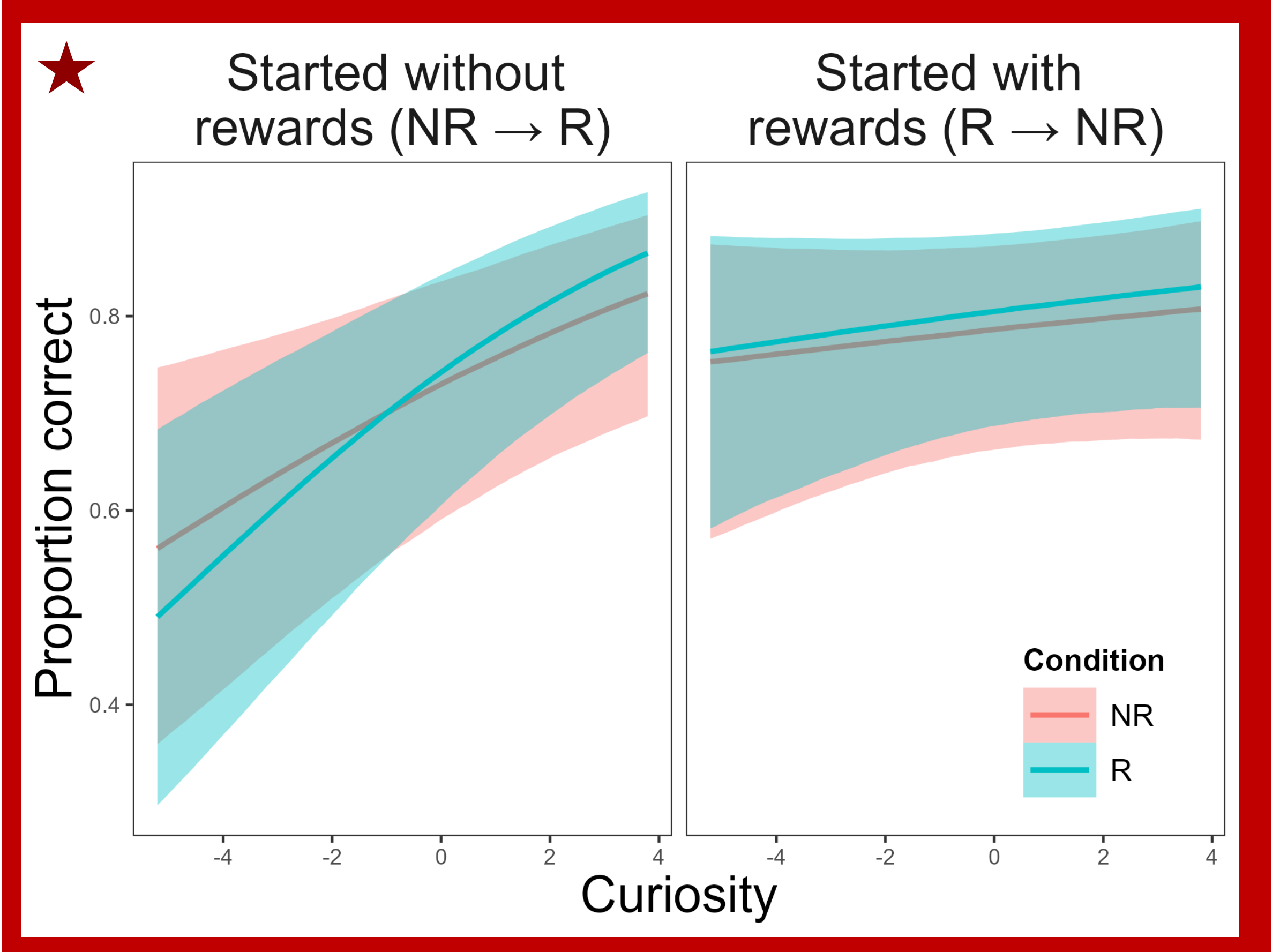
Reward (R) vs. No Reward (NR) Condition model



Reward only model

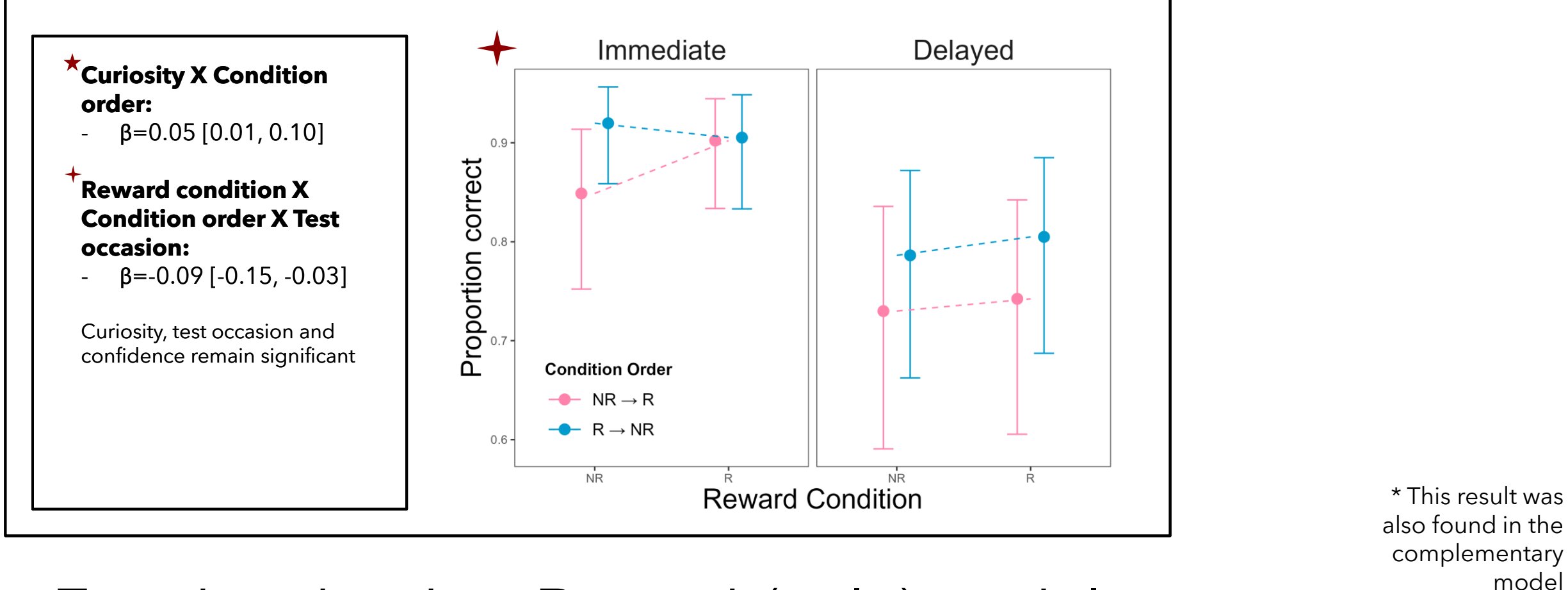


Curiosity and reward incentives boost learning... ...sometimes

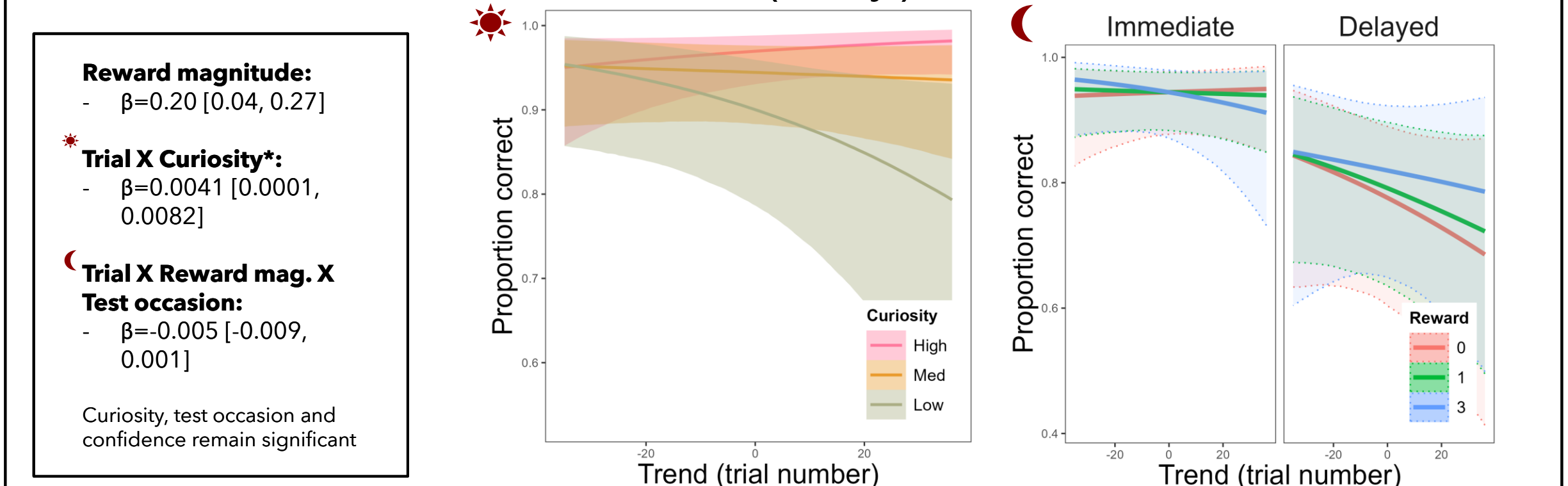


EXPLORATORY RESULTS

Trend and order: Condition model



Trend and order: Reward (only) model



DISCUSSION

STUDY

- The effects of intrinsic and extrinsic **motivations** on **memory** are complex and **dynamic**.
- **Curiosity** robustly and positively predicts **recall**.
 - **Reward** incentives did not seem to boost memory in our preregistered analyses.
- Effects** of curiosity and reward **change** when we consider **temporal dynamics**, i.e., trend and order effects.
- Still, curiosity and reward don't seem to interact directly, which may suggest they are differently processed in the brain.
 - Temporal dynamics do matter:
 - Being in a rewarded context from the beginning of the task immobilised the effect of curiosity completely, irrespective of receiving the reward or not.
 - Being rewarded only boosts memory for immediate recall, if you receive rewards after having had none before.
 - Curiosity becomes more effective after some time. This also counts for reward incentives, only in delayed recall performance.
 - In conclusion, **the brain seems to tune the reward system** to what is available. The **effect of future incentives** (curiosity or money) on memory **depends on what came before**.

EDUCATION

- Don't start with the extrinsic reward (grade, money, cake). Let curiosity do its work where it can. Immediately using rewards might even ruin curiosity's positive effect.
- Curiosity doesn't lose its effect over time, rewards do.
- Giving rewards can help in the short-term but will have drawbacks in the future.

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CONFIDENCE

- Confidence and curiosity have an inverted-U relationship
 - Tested using the two-line method by Uri & Leif (2014)
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